MILITARY SPECIFICATION

TELESCOPE: XM129

1. SCOPE

1.1 This specification covers Telescope XM129. A dual power arrangement permits the telescope to be used as an 8 or 12 power instrument. An articulated joint permits the objective tube to elevate coaxially with the weapon while the eyepiece remains stationary (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F8565556	Pressure Tester, Two-Station
F10555900	Telescope, XM129

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PACKAGING DATA SHEET

10555900 Packaging of Telescope, XM129

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabricaiton. - The telescope shall be manufactured in accordance with Drawing F10555900 and drawings pertaining thereto.

3.2 General specification. - The contractor shall be responsible for compliance with the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship
(f) Soldering electrical connections

3.3 Environmental.

3.3.1 Storage temperature. - The telescope shall show no evidence of physical failure such as cement separation, glass breakage, or casting failure after having been exposed and thermally stabilized at temperatures of -80° and +160° Fahrenheit (F).

3.3.2 Extreme operating temperature. - The telescope shall meet the applicable requirements of 3.8 while exposed and thermally stabilized at -40° and +150°F.

3.3.3 Vibration. - The telescope shall show no evidence of physical failure due to being vibrated successively along three (3) mutually perpendicular axes. Vibration shall be a simple harmonic motion, the frequency varying either linearly or logarithmically from 5 to 55 to 5 CPS in five (5) minute cycles for a total of 15 minutes along each of the axes. The amplitude of vibration shall be maintained constant at 1/64 inch (1/32 inch total excursion). Following the vibration test the telescope shall meet all subsequent requirements.

3.4 Cleanliness. - The telescope in each of the two power positions shall meet the cleanliness requirement of MIL-O-13830 when viewed through the eyepiece. The telescope shall show no more than five particles of dirt 0.2 mil maximum real diameter in the outer 35 degrees, and four particles of dirt 0.05 mil maximum real diameter in the central 25 degrees of the apparent field. There shall be no other dirt which shall adversely affect observation ans image quality.

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3.5 Sealing. - The telescope shall show no evidence of leakage in excess of 0.2 pound per square inch gage when subjected to an internal pressure of dry nitrogen of 5 ± 0.10 pounds per square inch gage (psig) for a minimum of 2 hours.

3.6 Orientation. - With the geometric axis of the objective tube (front tube) in a horizontal plane as established by its cylindrical mounting surfaces; the locating key at the six o'clock position and the center of the eyepiece hanger bolt hole shall be positioned 9.776 ± 0.10 inches below the plane of the geometric axis of the objective tube.

3.6.1 Eight power (8X position). - The requirements of 3.6.1.1 through 3.6.1.12.3 shall be met with the power selector lever at the 8X position.

3.6.1.1 Collimation. - With the geometric axis of the missile reticle in coincidence with the geometric axis of the telescope, the following requirements shall be met:

3.6.1.1.1 Conventional reticle. - The boresight cross of the conventional reticle shall be 26.6 mils ± 0.25 mil avove thegeometric axis of the telescope and be in coincidence with a plumb line passing through the geometric axis.

3.6.1.1.2 Missile reticle. - The boresight collimating square of the missile reticle shall be set at the 12 o'clock position.

3.6.1.2 Missile reticle movement. - From the position established in 3.6.1.1 rotation of the vertical boresight knob shall move the line of sight a minimum of 3.5 mils in depression and 3.5 mils in elevation. Rotation of the horizontal boresight knob shall move the line of sight a minimum of 5.5 mils to the left and 5.5 mils to the right.

3.6.1.3 Conventional reticle movement. - From the position established in 3.6.1.1.1 rotation of the vertical boresight knob shall move the line of sight a minimum of 6 mils in elevation and 9 mils in depression. Rotation of the horizontal boresight knob shall move the line of sight a minimum of 7.5 mils to the left and 7.5 mils to the right.

3.6.1.4 Backlash. - Backlash in any boresight mechanism shall not exceed 0.1 mil.

3.6.1.5 Reticle wander. - The image of the reticle boresight mark shall not deviate by more than one degree from a fixed reference in an auxiliary telescope secured to the eyepiece of the telescope when the telescope is articulated through its full range (from stop to stop).

3.6.1.6 Reticle lean. - The angular rotation of the missile reticle with respect to a reference in an auxiliary telescope secured to the eye-piece shall not exceed one degree when the telescope is articulated through its full range.

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3.6.1.7 Image tilt. - The image of a vertical target shall be vertical within one degree.

3.6.1.8 Resolution (8X power). - The resolution on the optical axis shall be 6.0 seconds of arc or less. Resolution consists of detectable line structure and proper line count in all four meridians within .25 diopter measured at the eyepiece.

3.6.1.9 Eyepiece focus. - Eyepiece focus shall be adjustable over the range of plus or minus 4 diopters as read on the eyepiece scale. The diopter scale shall indicate zero diopter within 0.25 diopter when the image of the conventional reticle is brought into best focus by eyepiece adjustment when viewing with a dioptometer set for zero diopter.

3.6.1.10 Reticle accuracy. - The angular value for reticle graduation between the zero and maximum horizontal range line shall be as indicated on the reticle drawing within 0.2 mil.

3.6.1.11 Patallax. - The total parallax between either reticle and the target image at the center of the field of view shall not exceed 0.1 roil, when viewing a target between 600 meters and infinity, with the parallax adjustment set at the midpoint of total travel. (Parallax is nominally zero at 1200 meters.) Parallax between retitles shall not exceed 0.1 mil.

3.6.1.12 Parallelsim of reticle and image. - With the telescope positioned in accordance with 3.6 the following requirements shall be met:

3.6.1.12.1 Missile reticle. - The horizontal reticle line of the missile reticle shall be parallel to the image of a horizontal target line within 30 minutes of arc.

3.6.1.12.3 Conventional reticle.. - The representative 1600 meter horizontal reticle line (total area of dashes and spaces) of the conventional reticle shall be parallel to the image of a horizontal target line within 30 minutes of arc.

3.6.1.12.3 Variation of settings. - The angular difference between the horizontal reticle line of the missile reticle and the 1600 meter line of the conventional reticle shall not exceed 30 minutes of arc.

3.6.2 Twelve power (12X position). - The requirements of 3.6.2.1 through 3.6.2.3 shall be met with the power selector lever at the 12X position.

3.6.2.1 Resolution (s2X power). - The resolution on the optical axis shall be 6.0 seconds of arc or less. Resolution consists of detectable line structure and proper line count in all 4 meridians within .25 diopter measured at the eyepiece.

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3.6.2.2 Focal plane shift. - The image of the missile reticle shall remain in sharp focus within .25 diopter when the power selector lever is changed from the 8x position to the 12X position.

3.6.2.3 Reticle image shift at eyepiece. - The angular change in the missile reticle line opf sight, when shifting from the 8x system to the 12X system, shall not exceed 5 degrees relative to an auxiliary telescope secured to eyepiece of the telescope.

3.7 Operability.

3.7.1 Diopter knob disengagement. - The force required to disengage the diopter knob from its normal position shall be 4 ± 2 pounds at standard ambient temperature (60° to 90°F) and shall be 6 ± 4 pounds at -40° and +150°F.

3.7.2 Diopter knob rotation. - The torque required to rotate the disengaged diopter knob shall be 2 ± 1 inch-pounds at -40" and +150°F.

3.7.3 Fo;ter tranmsfer lever. - The filter transfer lever shall be operable and shall firmly seat against the stops at -40° and +150°F.

3.7.4 Articulation torque. - With the telescope on its right or left side, unrestricted, the torque required to rotate the objective tube about the articulated joint through full movement shall not exceed 50 foot pounds at a standard ambient temperature between 60° and 90°F, and 75 foot pounds at -40° and +150°F.

3.7.5 Boresight knob disengagement. - The force required to disengage the boresight knobs from their normal position shall be 4 ± 2 pounds at standard ambient temperature and shall be 6 ± 4 pounds at -40° and +150°F.

3.7.6 Boresight knob rotation. - The torque required to rotate the disengaged boresight knobs shall be 3 ± 1 inch-pounds at standard ambient temperaute, and shall be 4 ± 3 inch-pounds at -40° and +150°F.

3.7.7 Power slector lever. - The power selector lever shall be operable and shall firmly seat at -40° and +150°F.

3.7.8 Lanp housing assemvlies. - The lamp housing assemblies shall be mechanically operable at standard ambient temperature and at extreme temperatures of 3.3.2 to facilitate replacement of reticle lamp.

3.8 Electrical.

3.8.1 Reticle illumination. - With the reticle light for each reticle energized by a 24 volt power source, the illuminated reticle shall appear clearly defined when observed in a darkened area when the filters are in the clear position.

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3.8.2 Circuitry. - The circuitry of the telescope shall be in accordance with the applicable electrical schematic drawings.

3.8.3 Dielectric strength. - There shall be no puncture of insulation, no sparkover, no decrease in dielectric strength, when the insulated area of the circuitry is subjected to an alternating current (at) root mean square (rms) potential of 500 volts at a frequency of 60 cps for a period of 15 ± 5 seconds.

3.8.4 Insulation resistance. - Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct current (dc) applied

3.9 interior atmosphere. - There interior atmosphere of the accepted telescope shall contain nitrogen having a dew point at least as low as -25°F and the internal pressure of the telescope shall be between 0.10 and 0.25 psig.

3.10 Workmanship. - The workmanship requirements of MIL-F-13926 shall be adhered to. Exterior protective surface finishes shall be free of voids which may affect the protective qualities of the finish or appearance. Optical surfaces shall be free of scratches and blemishes which may affect their optical qualities or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production)appproval. - The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 telescopes plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes,

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and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Governmetn is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materiasls, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified y the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% defective		
101. Vibration	3.3.3	4.6.3
102. Cleanliness	3.4	4.6.4
103. Sealing	3.5	4.6.5
104. Interior atmosphere	3.9	4.6.6
105. Collimation, conventional reticle	3.6.1.1.1	4.6.7.1.1
106. Collimation, missile reticle	3.6.1.1.2	4.6.7.1.2
107. Missile reticle movement	3.6.1.2	4.6.7.1.3
108. Conventional reticle movement	3.6.1.3	4.6.7.1.4
109. Backlash, missile reticle	3.6.1.4	4.6.7.1.5
110. Backlash, conventional reticle	3.6.1.4	4.6.7.1.6
111. Reticle wander	3.6.1.5	4.6.7.1.7
112. Reticle lean	3.6.1.6	4.6.7.1.8
113. Image tilt	3.6.1.7	4.6.7.1.9
114. Resolution, 8 power	3.6.1.8	4.6.7.1.10
115. Eyepiece focus	3.6.1.9	4.6.7.1.11
116. Reticle accuracy	3.6.1.10	4.6.7.1.12
117. Parallax	3.6.1.11	4.6.7.1.13
118. Parallelism of missile reticle and image	3.6.1.12.1	4.6.7.1.14
119. Parallelism of conventional reticle and image	3.6.1.12.2	4.6.7.1.15
120. Variation of settings	3.6.1.12.3	4.6.7.1.16
121. Resolution, 12X	3.6.2.1	4.6.7.2.1
122. Focal plane shift, 12X	3.6.2.2	4.6.7.2.2
123. Reticle image shift at eyepiece, 12X	3.6.2.3	4.6.7.2.3
124. Diopter knob disengagement	3.7.1	4.6.8
125. Diopter knob rotation	3.7.2	4.6.9
126. Filter transfer lever	3.7.3	4.6.10
127. Articulation torque	3.7.4	4.6.11
128. Boresight knob, disengagement	3.7.5	4.6.12
129. Boresight knob, rotation	3.7.6	4.6.13
130. Lamp housing assemblies	3.7.8	4.6.15
131. Reticle illumination	3.8.1	4.6.16
132. Circuitry	3.8.2	4.6.17
133. Dielectric strength	3.8.3	4.6.18
134. Insulation resistance	3.8.4	4.6.19
135. Workmanship	3.10	Visual-Tactile
MINOR: NONE		

NOTE: The tests of Table I shall be conducted at a temperature between +60° and +90°F.

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4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defedtive units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General. - One telescope shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawing - visual
302.	General specification	3.2	MIL-F-13926 - visual

4.4.2 Environmental. - Three telescopes shall be selected at random as samples from each 50 produced, or from each month's production, whichever occures first. Each sample shall meet the requirements and tests in Table III.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperatures (-80° and +160°F)	3.3.1	4.6.1
304.	Diopter knob disengagement	3.3.2, 3.7.1	4.6.2, 4.6.8
305.	Diopter knob rotation	3.3.2, 3.7.2	4.6.2, 4.6.9
306.	Filter transfer lever	3.3.2, 3.7.3	4.6.2, 4.6.10
307.	Articulation torque	3.3.2, 3.7.4	4.6.2, 4.6.11
308.	Boresight knob disengagement	3.3.2, 3.7.5	4.6.2, 4.6.12
309.	Boresight knob rotation	3.3.2, 3.7.6	4.6.2, 4.6.13
310.	Power selector lever	3.3.2, 3.7.7	4.6.2, 4.6.14
311.	Lamp housing assemblies	3.3.2, 3.7.8	4.6.2, 4.6.15

NOTE: The tests for Characteristics 304 through 311 inclusive shall be conducted at -40° and +150°F.

4.4.3 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

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4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Sealing. - The pressure testing equipment to test sealing of the telescope shall conform to the internal pressure test requirements of Sheet 1 of Drawing F8565556, Pressure Tester-Two Station.

4.5.2.3.2 Orientaion of telescope. - The testing equipment utilized to mount and orient the telescope for the accuracy tests shall permit mounting the geometric axis of the objective tube (front tube) in a horizontal plane within ten seconds, the locating key at the six o'clock position within ten seconds, and the center of the eyepiece hanger bolt hole positioned 9.776 ± 0.010 inches below the plane of the objective tube. The testing equipment shall also provide for articulation of the eyepiece housing assembly to any position in elevation between its stops.

4.5.2.3.2.1 Collimation, missile reticle movement, conventional reticle movement, backlash, reticle accuracy and parallax. - The testing equipment used for these tests shall be accurate within eighteen seconds.

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4.5.2.3.2.2 Reticle wander, reticle lean, image tilt and reticle image shift. - The testing equipment used for these tests shall be accurate within six minutes.

4.5.2.3.3 Force requirements (disengagement). - Testing equipment for measuring direct force (pull) with a minimum range of zero to twelve pounds, the accuracy of which shall be plus or minus two ounces throughout the entire zone of excursion, shall be utilized. The testing equipmetn shall be equipped with suitable adapters to accommodate testing of the diopter knob and boresight knobs.

4.5.2.3.4 Torque requirements. - A torque wrench tester with a minimum range from zero to tem inch pounds, the accuracy of which shall be plus or minus two inch ounces throughout the entire zone of excursion and a torque wrench type tester with a minimum range from zero to one hundred foot pounds, the accuracy of which shall be plus or minus one-half foot pound throughout the entire zone of excursion, shall be utilized. The testing equipment shall be equipped with suitable adapters to accommodate testing of the diopter knob, objective tube and the boresight knobs.

4.5.2.3.5 Set-up and calibration. - Rigidity, stability, ease of maintenance, tenance, and adaptability of set-up and calibration equipment with complete instructions are also prime requirements. All set-up and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment.

4.6 Test methods and procedures.

4.6.1 Storage temperature. - The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.1 for a minimum of 5 hours. After each temperature exposure the telescope shall be stabilized in an ambient temperature between 60° and 90° Fahrenheit (F) and subjected to the visual tactile, and audible examination of 4.6.3.1. The telescope shall conform to the requirements of 3.3.1. Caution shall be exercised during this test to avoid subjecting the telescope to thermal shock.

4.6.2 Extreme operating temperature. - The testin equipment utilized in this test. shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3,3.2. Th telescope shall be exposed to each temperature extreme specified in 3.3.2 for a period of 5 hours. At the end of each 5 hour period, and while still at the temperature extreme, the telescope shall be subjected to the applicable test of 4.6.8 through 4.6.15 inclusive. The telescope shall conform to the requirements of 5.3.2. Caution shall be exercised during this test to avoid subjecting the telescope to thermal shock.

4.6.3 Vibration. - Vibrating equipment capable of providing the amplitude and frequency specified in 3.3.3 shall be utilized for this test. The vibrating equipment shall also be capable of vibrating the telescope along 3 mutually perpendicular axes while the telescope is maintained in its using position, as depicted in the lower view on sheets 1 and 2 of Drawing F10555900.

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The telescope shall be horizontally mounted on the vibrating table by means of an adapter. This adapter shall immobilize the objective tube assembly, and the offset tube assembly, so that the articulating joint assembly is immobilized in its midpoint of the articulating range. The telescope shall be vibrated in accordance with Procedure IX' Part 1 of MIL-STD-810, except that the frequency, frequency cycling, and amplitude of the vibration shall be as specified in 3.3.3. Upon completion of the vibration cycling the telescope shall be subjected to the tests specified in 4.6.3.1.

4.6.3.1 Examination. - Subsequent to vibration the telescope shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine conformance with 3.3.3, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall also be conducted to determine looseness or improper functioning of moving parts such as the articulated joint, diopter adjustment, boresight knobs, power selector lever, etc. An audible examination shall be made of the telescope by shaking it and listening for loose internal parts or broken optical components, such as lenses, prisms, retitles, ets. The electrical circuit shall be inspected for damage by energizing each reticle light (see 4.6.7) and conducting a visual examination for malfunction such as: no reticle illumination, flicker, intermittent illumination, etc. Any one of the possible defects defined above shall be cause to consider a telescope defective. Telescopes which have passed the above examinations shall be subjected to the tests specified in 4.6.4 through 4.6.19 inclusive to determine overall compliance with 3.3.3.

4.6.4 Cleanliness. - The optical surfaces in each of the two power positions, shall be inspected for cleanliness in accordance with MIL-O-13830. The telescope shall conform to the cleanliness requirements of 3.4. Dirt particles which interfere with the legibility or accuracy of the reticle patterns shall be cause for rejection.

4.6.5 Sealing. - The internal pressurizing of the telescope (telescope assembly and reticle projector assembly), to test sealing shall be performed with special testing equipment conforming to 4.5.2.3.1. The pressure shall be applied through the applicable valve in the telescope assembly and reticle projector assembly. Pressurize each assembly with dry nitrogen gas to 5.0 ± 0.10 psig in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. The dry nitrogen gas shall have a dew point at least as low as minus 25°F. The applied pressure in each assembly (telescope and reticle projector) shall be retained for a minimum of 2 hours. Within the first 60 minutes of this 2 hour time period, the telescope eyepiece housing assembly shall be elevated through a minimum of 3 full excursions of the articulated joint assembly, the power selector lever shall be repositioned a minimum of 3 times, the diopter adjustment shall be rotated through a minimum of 3 full excursions (cw stop-to ccw stop-to cw stop), of its rotational range and each elevation and deflection boresight knob shall be rotated through

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a minimum of 3 full excursions (cw stop-to ccw-to cw stop) of their rotational range. Any evidence of leakage in the telescope assembly or reticle projector assembly, at the end of the 2 hour time period, shall not exceed the tolerance specified in 3.5, as measure on the pressure gauges of the testing equipment.

4.6.6 Interior atmosphere. - Upon satisfactory completion of the sealing test the telescope shall be flushed with the dry nitrogen gas. The flushing shall be accomplished by opening the purging vent on the objective tube and allowing the nitrogen gas to flow through the telescope, until gas bled from the telescope conforms to the dew point requirements of 3.9, as measured with a dew point tester. Close the purging vent and reduce the internal pressure of the telescope to the value specified in 3.9, as measured with sensitive pressure testing equipment. The same procedure for flushing shall be applied to the reticle projector assembly by opening the purging vent on the housing assembly containing the elevation knob.

4.6.7 Orientation. - Orientation shall be accomplished by means of special testing equipment in compliance with 4.5.2.3.2, placed on a surface plate or table that has been leveled by means of a precision (10 seconds) bench level. The end of the objective tube assembly shall be positioned toward a target for measuring the projected reticle patterns of the telescope. Unless otherwise specified, optical accuracy of the telescope shall be measured on a collimator projector target set to a focus of 1200 meters ±100 meters. The optical axis of the collimator projector shall be coincident to the geometric axis of the telescope. A reticle switch selector to select either of the two reticle patterns for projection into the optical path of the telescope, and a 24 volt dc power source to provide the power to energize the telescope will be used in conjunction with the testing equipment specified in 4.5.2.3.2. The telescope shall be positioned as specified in 3.6 unless otherwise specified herein.

4.6.7.1 Eight power (8x) system. - The following tests shall be conducted with the power selector lever at the 8x position.

4.6.7.1.1 Collimation, conventional reticle. - Actuate the reticle selector switch to illuminate the missile reticle. Position the boresight cross of the missile reticle (intersecting point of the vertical and horizontal lines) coincident with the intersecting point of the vertical and horizontal target line image on the collimator projector target, by means of the missile reticle boresight knobs. A reference mark shall be made on the dial control plate in true alignment with a reference mark on the housing which will serve as a zero reference point of midpoint position for the missile boresight knobs. Actuate the reticle switch to illuminate the conventional reticle. Position the boresight cross of the conventional reticle (intersecting point of the vertical and horizontal lines) coincident with the image of a target reference point located on the vertical target line 26.6 mils above the horizontal target line, by means of the conventional reticle boresight knobs. Slip

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the conventional reticle knob scales to their zero reference position. The missile and conventional retitles shall conform to the requirements of 3.6.1.1 and 3.6.1.1.1.

4.6.7.1.2 Collimation missile reticle. - Actuate the reticle selector switch to illuminate the missile reticle. Position the boresight cross of the reticle as in 4.6.7.1.1. Observe the position of the boresight collimating square in relation to the image of the vertical line of the target reticle. The boresight collimating square of the missile reticle shall conform to the requirements of 3.6.1.1.2.

4.6.7.1.3 Missile reticle movement. - Conduct this test in sequence to the test in 4.6.7.1.2. From the position established in 4.6.7.1.1 rotate the missile reticle horizontal and vertical boresight knobs, in turn, to their clockwise stop, then to their counterclockwise stop. During this excursion of the boresight knobs, the line of sight through the boresight cross of the missile reticle shall move the number of mils specified in 3.6.1.2, as measured on the target.

4.6.7.1.4 Conventional reticle movement. - Actuate the reticle selector switch to illuminate the conventional reticle. From the position established in 4.6.7.1.1 rotate the conventional reticle horizontal and vertical boresight knobs, in turn, to their clockwise stop, then to their counterclockwise stop. During this excursion of the boresight knobs, the line of sight through boresight cross of the conventional reticle shall move the number of mils specified in 4.6.1.3.

4.6.7.1.5 Backlash, missile reticle. - Actuate the reticle selector switch to illuminate the missile reticle. With backlash excluded, rotate the horizontal boresight knob in a continuous clockwise direction until the vertical reticle line is in coincidence with the vertical target line. A mark shall be made on the dial control in true alignment with a mark on the housing which shall serve as a reference point. Proceed in the same clockwise direction for at least one half turn of the knob, then reverse rotation of the knob manually in a continuous counterclockwise direction (with backlash included) until the reference point is reached without overtravel. Observe the position of the vertical reticle line with reference to the vertical target line and read the mil scale of the target. Any difference between the vertical reticle line and the vertical target line is the amount of backlash. Repeat this operation with the vertical boresight knob, using the horizontal line of the scaled target to measure backlash. Backlash for each missile reticle boresight mechanism shall not exceed the limit specified in 3.6.1.4.

4.6.7.1.6 Backlash, conventional reticle. - Actuate the reticle selector switch to illuminate the conventional reticle. With backlash excluded, rotate the deflection boresight knob in a continuous clockwise direction until the verticle reticle line of the boresight cross is in coincidence with the vertical target line. Slip the knob mil scale to its zero position. Proceed to rotate the knob in the same clockwise direction

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for at least one-half turn, then reverse rotation of the knob in a continuous counterclockwise direction (with backlash included) until the vertical reticle line of the boresight cross is coincident with the vertical target line, without overtravel. Read the mil scale on the knob. Any difference between the zero position and the post rotation reading on the knob mil scale is the amount of backlaxh. Repeat this operation with the elevation boresight knob using the knob scale to measure backlash. Backlash for each conventional reticle boresight mechanism shall not exceed the limit specified in 3.6.1.4.

NOTE: During these operations (4.6.7. 1.5 and 4.6.7.1.6), if overtravel occurs, the test shall be nullified and the procedures specified for backlash shall again be performed.

4.6.7.1.7 Reticle wander. - Actuate the reticle selector switch to illuminate the missile reticle. Articulate the telescope eyepiece housing assembly to the position specified in 3.6. Affix an auxiliary telescope to the eyepiece of the telescope under test.The reticle pattern of the auxiliary telescope shall contain a vertical and horizontal reticle line, and shall also contain etched data capable of measuring 1 degree of reticle wander. The auxiliary telescope shall be of at least 3 power (3x). Obtain coincidence of the intersecting point of the auxiliary telescope reticle lines with the intersecting point of the horizontal and vertical reticle lines (boresight point) of the missile reticle, and plumb the vertical reticle line of the auxiliary telescope. The image of the telescope reticle boresight point shall not deviate in excess of 1° from the intersecting point of the auxiliary telescope reticle, when the articulating telescope is actuated as specified in 3.6.1.5. Reticle wander shall be measured on the auxiliary telescope reticle with reference to the telescope reticle.

4.6.7.1.8 Reticle lean. - The test shall be performed with the aid of the auxiliary telescope positioned as specified in 4.6.7.1.7. The auxiliary telescope reticle shall be capable of measuring 1° reticle lean. Adjust the auxiliary telescope vertical line coincident to the vertical reticle line of the telescope missile reticle. Elevate and depress the telescope eyepiece housing as specified in 3.6.1.6. The image of the articulated telescope's vertical reticle line shall remain parallel to the auxiliary telescope's vertical reticle line within the tolerance specified in 3.6.1.6 during this actuation. Reticle lean shall be measured on the auxiliary telescope reticle with reference to the telescope's vertical reticle line.

4.6.7.1.9 Image tilts. - The test shall be performed with the aid of the auxiliary telescope positioned as specified in 4.6.7.1.7. Adjust the auxiliary telescope vertical reticle line plumb. Obtain coincidence of the intersecting point of the auxiliary telescope reticle lines with the intersecting point of the target vertical and horizontal lines. The image of the vertical target line shall be vertical within the tolerance specified in 3.6.1.7. Image tilt shall be measured on the target with reference to the vertical target line and the auxiliary telescope vertical reticle line.

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4.6.7.1.10 Resolution 8 power. - This test shall be performed utilizing an auxiliary telescope with a magnification of at least 3 power (3X), a resolving power chart in a collimator projector representing the angular subtense for the seconds of arc specified for resolution, and a supporting medium to receive this test equipment as well as the telescope to be tested. The resolving power chart shall contain four line sets as shown in Figure 1 of MIL-O-13830. The test method and procedure shall be in compliance with the "Resolution Test" specified in MIL-O-13830. The limit of resolution shall be within the value specified in 3.6.1.8 as determined by viewing through the auxiliary telescope.

4.6.7.1.11 Eyepiece focus. - Actuate the reticle selector switch to illuminate the coventional reticle. The test shall be made with the aid of a calibrated dioptometer having a magnification of at least 3 power (3X), and a diopter measuring range of at least plus or minus 4 diopters. Adjust the dioptometer eyepiece for best focus of the dioptometer reticle, and set the dioptometer diopter scale to zero. Position the dioptometer at the eyepiece of the telescope. While viewing through the dioptometer, obtabest focus of the conventional reticle pattern by rotation of the eyepiece assembly of the telescope. Rotation of the eyepiece assembly to measure the accuracy of the zero setting of the diopter scale shall be made with backlash removed for directionof travel. The telescope reticle pattern shall be in best focus within the zero diopter tolerance specified in 3.6.1.9, as measured on the telescope diopter scale. Remove the dioptometer at the telescope eyepiece. Rotate the eyepiece assembly of the telescope to the zero diopter position as indicated on the diopter scale. From this zer diopter position rotate the eyepiece assembly to its clockwise stop and note the diopter scale setting, then rotate the eyepiece assembly to its counterclockwise stop and note the diopter scale. The eyepiece focus adjustment range shall conform to the requirements specified in 3.6.1.9, as measured on the diopter scale of the telescope.

4.6.7.1.12 Reticle accuracy. - Actuate the reticle selector switch to illuminate the conventional reticle. The test for reticle accuracy shall be performed in accordance with MIL-O-13830. The angular subtense of the reticle pattern between the zero and maximum horizontal range line shall not exceed the tolerance specified in 3.6.1.10, as measured on the target.

4.6.7.1.13 Parallax. - The test shall be performed with the diopter knob of the telescope set for best focus of the telescope reticle, and the parallax adjustment set as specified. The line of sight through each reticle (conventional and missile) in the telescope shall be directed to calibrated targets located at 600 meters, 1200 meters and infinity. Parallax is any apparent movement of the reticle lines in relation to the target image when the observer's head is moved from side to side or up and down. Parallax at the center of the field of view for each reticle (conventional and missile) shall be recorded. Parallax between each reticle and the target image, and parallax between retitles (conventional and missile) shall not exceed the applicable tolerance specified in 3.6.1.11. Upon completion of this test return the target collimator to the focus established in 4.6.7.

4.6.7.1.14 Parallelism of missile reticle and image. - Actuate the reticle selector switch to illuminate the missile reticle. Sight through

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the eyepiece of the telescope and obtain coincidence of one end of the telescope's horizontal reticle line with that of the horizontal target line image. Then observe for coincidence at the opposite end of the horizontal line. The horizontal reticle line shall be parallel to horizontal target line image within the tolerance specified in 3.6.1.12.1, as measured on the target. Deviation shall be noted and recorded.

4.6.7.1.15 Parallelism of conventional reticle and image. - Actuate the reticle selector switch to illuminate the conventional reticle. Sight through the eyepiece of the telescope and obtain coincidence of one end of the telescope's 1600 meter horizontal reticle line (total length of dashes and spaces) with that of a horizontal target line. Then observe for coincidence at the opposite end of the reticle line. The horizontal line shall be parallel to the horizontal target line within the tolerance specified in 3.6.1.12.2, as measured on the target. Deviation shall be noted and recorded.

4.6.7.1.16 Variation of settings. - The angular difference between the recorded deviation for the missile reticle and conventional reticle obtained in 4.6.7.1.14 and 4.6.7.1.15 shall not exceed the tolerance specified in 3.6.1.12,3.

4.6.7.2 Twelve power (12X) system. - The following tests shall be conducted with the power selector lever at the 12x position, except as otherwise specified, and the telescope eyepiece housing assembly positioned as specified in 3.6.

4.6.7.2.1 Resolution 12 power. - Perform this test as specified in 4.6.7.1.10. Resolution of the 12X system shall conform to the requirements of 3.6.2.1.

4.6.7.2.2 Focal plane shift. - The test shall be made with the aid of a calibrated dioptometer with a magnification of at least 3 power (3X). Adjust the dioptometer eyepiece for best focus of the dioptometer reticle and set the dioptometer diopter scale to zero. Set the diopter scale of the telescope under test to zero diopter. Position the dioptometer behind the eyepiece assenmbly. Move the power selector lever of the telescope to the 8X position. Obtain best focus of the telescope missile reticle by adjustment of the dioptometer diopter scale. Note the diopter setting on the dioptometer diopter scale. Move thje power selector lever of the telescope to the 12X position. Again obtain best focus of the missile reticle by adjustment of the dioptometer diopter scale. Note the diopter setting on the dioptometer diopter scale. Any difference between the dioptometer diopter scale setting for the 8X system and the 12X system shall not exceed the tolerance specified in 3.6.2.2.

4.6.7.2.3 Reticle image shift at eyepiece. - Actuate the reticle selector switch to illuminate the missile reticle. Move the power selector lever to the 8X position. Affix an auxiliary telescope to the eyepiece of the telescope under test. The reticle pattern of the auxiliary telescope

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shall contain a vertical and horizontal reticle line, and shall also contain etch data capable of measuring 5 degrees of reticle image shift. The auxiliary telescope shall be of at least 3 power (3X). Obtain coincidence of the intersecting point of the auxiliary telescope reticle lines with the intersecting point of the horizontal and vertical reticle lines (boresight point) of the missile reticle, and plumb the vertical reticle line of th auxiliary telescope. The image of the telescope reticle boresight point shall not deviate in excess of the tolerance specified in 3.6.2.3 when the power selector lever is moved to the 12X position. Reticle image shift shall be measured on the auxiliary telescope reticle with reference to the telescope reticle.

4.6.8 Diopter knob disengagement. - Testing equipmetn conforming to the requirements of 4.5.2.3.3 shall be utilized for this test. The force The force (pull) required to disengage the diopter knob from its normal position shall be within the values specified in 3.7.1 for the temperatures indicated.

4.6.9 Diopter knob rotation. - Testing equipment conforming to the requirements of 4.5.2.3.4 shall be utilized for this test. The running torque required to rotate the diopter knob throughout its range of movement, disengaged shall be within the values specified in 3.7.2 for the temperatures indicated.

4.6.10 Filter transfer lever. - When the telescope is thermally stabilized at -40° and +150°F the filter transfer lever shall be subjected to a tactile inspection. The filter transfer -lever shall conform to the requirements of 3.7.3 at each extreme temperature.

4.6.11 Articulation, torque. - Testing equipment conforming to the requirements of 4.5.2.3.4 shall be utilized for this test. The adapter shall be used with the torque measuring equipment to accommodate the objective tube assembly at the articulated joint. Position the telescope as specified in 3.7.4. Torque shall be measured while rotating the objective tube through its full movement about the articulated joint with the eyepiece housing assembly immobilized. The maximum running torque required to rotate the objective tube shall not exceed the value specified in 3.7.4 for the temperature indicated.

4.6.12 Boresight knob disengagement. - Testing equipment conforming to the requirements of 4.5.2.3.3 shall be utilized for this test. The force required to disengage each boresight knob from its normal position shall be within the values specified in 3.7.5 for the temperatures indicated.

4.6.13 Boresight knob rotation. - Test equipment conforming to the requirements of 4.5.2.3.4 shall be utilized for this test. The running torque required to rotate each boresight throughout its range of movement, disengaged, shall be within the values specified in 3.7.6 for the temperatures indicated.

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4.6.14 Power selector lever. - When the telescope is thermally stabilized at -40° and +150° F, the power selector lever shall be subjected to a tactile inspection. The power selector lever shall conform to the requirements of 3.7.7 at each extreme temperature.

4.6.15 Lamp housing assemblies. - The lamp housing assemblies shall be subjected to a tactile examination. The missile and conventional reticle lamp housing assemblies shall conform to the requirements of 3.7.8 at the temperatures indicated.

4.6.16 Reticle illumination. - The illiuminated reticle etchings (conventional and missile) shall be clearly defined when visually examined against a darkened background. Should the brightness of a reticle defect be greater than the brightness of a reticle line, the defect shall be cause for rejection. Opaque areas shall be free of pinpoint holes, cracks or chips. The conventional and missile retitles shall conform to the requirements of 3.8.1.

4.6.17 Circuitry. - A continuity check of the telescope circuits shall be made using standard measuring equipment to determine conformance with the requirements of 3.8.2.

4.6.18 Dielectric strength. - The telescope shall be tested in accordance with Method 301 of MIL-STD-202, utilizing standard measuring equipment to determine conformance with the requirements of 3.8.3. All All lamps shall be removed for this test.

4.6.19 Insulation resistance. - Immediately after satisfactorily withstanding the dielectric strength test, the telescope shall be subjected to the insulation resistance test in accordance with Method 302, Test Condition B of MIL-STD-202. Measurements between isolated points and ground shall be made utilizing standard measuring equipment to determine conformance with the requirements of 3.8.4. All lamps and ground connection shall be removed for this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 10555900. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The Telescope XM129 is primarily intended as the direct fire telescope for the Gun, Tank: M60A1E2.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

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(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

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Preparing activity:

Army - MU

project No. 1240-A060

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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